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A review of rabies burden, diagnostic gaps, and control strategies in Rural India: Focus on Madhepura, Bihar

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Abstract- Rabies remains a deadly yet entirely preventable zoonotic disease that disproportionately affects low-resource regions, particularly in developing countries like India. Despite global efforts and available prophylactic measures, the disease continues to claim thousands of lives annually-most of them in rural areas where healthcare access is limited and public awareness is low. Madhepura district in Bihar exemplifies the systemic challenges that hinder effective rabies control, including inadequate diagnostic infrastructure, irregular vaccine availability, and insufficient public education. This review synthesizes global and national perspectives on rabies virology, epidemiology, and diagnostic innovations, while placing special emphasis on the ground-level realities of districts like Madhepura. It critically examines the applicability of novel molecular diagnostic techniques, such as Reverse Transcription Loop-Mediated Isothermal Amplification (RT-LAMP), which offer promising solutions for decentralized and field-based rabies detection in resource-poor environments. Furthermore, the review highlights the urgent need to operationalize a "One Health" approach-integrating human, animal, and environmental health interventions-to sustainably interrupt rabies transmission. By aligning scientific advancements with practical implementation strategies, this review aims to inform public health planning and policy decisions at the district level. The insights presented advocate for context-specific, community-driven, and multisectoral efforts to close the gap between policy and practice, ultimately contributing to India's goal of eliminating dog-mediated human rabies by 2030.

Key words: Rabies virus, Rhabdoviridae, post-exposure prophylaxis, One Health

INTRODUCTION

Rabies is persistently deadly viral disease that continues to present a major public health concern, especially in low- and middle-income countries. Caused by the Rabies virus, which belongs to the *Lyssavirus* genus of the Rhabdoviridae family, the disease is typically transmitted through bites or scratches from infected animals. Once the virus reaches the central nervous system, it causes severe inflammation of the brain (encephalitis), and death is almost inevitable after symptoms appear.[1,2] Although the disease is entirely preventable, it becomes untreatable once clinical signs develop, making early medical intervention-particularly post-exposure prophylaxis (PEP)-crucial for survival.

According to the World Health Organization (WHO)[3], rabies is responsible for around 59,000 human deaths annually, with more than 95% of these occurring in Asia and Africa. India shoulders the largest share of this burden, contributing an estimated 36% of the global rabies-related deaths.[4] Each year, the country reports more than 17 million dog bites, resulting in over 20,000 fatalities, largely due to

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bites from infected dogs. A significant proportion of these victims are children under 15, who often lack the knowledge or means to access timely and appropriate medical care.

Within India, the rabies burden is unevenly distributed. While urban regions benefit from better veterinary infrastructure, vaccination programs, and healthcare services, rural and marginalized areas continue to face disproportionately higher risks.[5] Madhepura, a district situated in Bihar's Kosi division, is a prime example of this inequality. The region is marked by widespread poverty, inadequate healthcare facilities, a large population of stray dogs, and low levels of public awareness-factors that together facilitate the persistent circulation of rabies.

The healthcare system in Madhepura is significantly under-resourced. Many Primary Health Centers (PHCs) do not have a regular supply of anti-rabies vaccines (ARVs) or rabies immunoglobulin (RIG), and there are no local facilities equipped for laboratory confirmation of rabies cases. Additionally, the absence of organized dog vaccination drives and population control measures allows the animal reservoir to grow unchecked. In such circumstances, it is common for victims to first approach traditional healers or adopt home remedies, delaying essential PEP due to lack of awareness, cultural preferences, or financial limitations.

Compounding these challenges is the absence of structured surveillance systems to monitor dog bites and rabies cases. This lack of reliable data hinders effective public health responses, as interventions are often delayed or poorly targeted in the absence of accurate epidemiological insight.

From a diagnostic perspective, the global scientific community has developed advanced tools such as Reverse Transcription Loop-Mediated Isothermal Amplification (RT-LAMP), which provides a fast, cost-effective, and highly sensitive alternative to standard RT-PCR and DFA techniques.[6,7] These methods are particularly suitable for use in resource-limited environments. However, despite their potential, such technologies have not yet been implemented in districts like Madhepura, where even basic laboratory infrastructure remains lacking.

Given these realities, the current review seeks to comprehensively examine the status of rabies as a continuing public health issue, with a specific emphasis on underserved rural districts like Madhepura. It delves into the disease's epidemiology, existing barriers to its control, diagnostic innovations, and the critical need for adopting a "One Health" approach-an integrated framework that connects human, animal, and environmental health to combat zoonotic diseases like rabies.

By aligning international and national insights with the on-the-ground challenges faced by vulnerable districts, this review aims to contribute to actionable strategies for improving rabies prevention and control. The broader objective is to support India's mission to eliminate dog-mediated human rabies by 2030, in line with WHO and World Organisation for Animal Health (OIE) global targets.

METHODS

This review article was developed through a structured and narrative synthesis of published and gray literature, national health reports, and global policy documents related to rabies epidemiology, prevention, diagnosis, and control. The methodology was designed to ensure comprehensive coverage of both scientific advances and public health challenges, with a specific focus on resource-limited districts such as Madhepura in Bihar, India.

Data Collection and Synthesis

Information from selected sources was categorized into thematic areas:

- Rabies epidemiology in rural Bihar (with a focus on Madhepura),
- Post-exposure prophylaxis (PEP) availability and access
- Diagnostics including DFA, RT-PCR, and RT-LAMP
- Health infrastructure and system-level gaps
- Public awareness and traditional health-seeking behavior
- One Health strategies and policy implementation

Epidemiological Burden in Madhepura and Similar Settings

Madhepura, located in the Kosi division of Bihar, typifies the high-risk profile for rabies persistence. Characterized by underdeveloped healthcare infrastructure, low literacy, and economic hardship, the region records a high frequency of dog bites. Data from community health records and national surveys indicate:

- Children (5-15 years) are the most affected demographic.

- A significant proportion of bite victims delay or avoid seeking modern medical treatment.
- Awareness about rabies as a fatal but preventable disease is remarkably low.
- Compliance with PEP protocols remains inconsistent, especially with limited availability of rabies immunoglobulin (RIG) and vaccines.[8,9]

These trends emphasize the need for context-specific interventions in awareness generation, access to vaccines, and reliable data collection systems.

Diagnostic Approaches to Rabies: Global Advances and Local Gaps

Traditionally, rabies diagnosis in animals relies on direct fluorescent antibody testing (DFA), while reverse transcription polymerase chain reaction (RT-PCR) is employed for ante-mortem diagnosis in humans.[10-12] However, both methods require sophisticated laboratory infrastructure, cold chain maintenance, and trained personnel-facilities that are absent in districts like Madhepura.

Recent advancements in molecular diagnostics offer alternative solutions. Among them, Reverse Transcription Loop-Mediated Isothermal Amplification (RT-LAMP) has emerged as a promising tool for resource-constrained regions. RT-LAMP is:

- Isothermal, requiring only basic heating equipment.
- Highly sensitive, capable of detecting viral RNA at fg-level concentrations.
- Field-adaptable, operable without high-end lab infrastructure.

Studies across endemic countries, including India and the Philippines, have validated RT-LAMP as a viable method for rapid, accurate rabies detection. Though these techniques have not yet been deployed in Madhepura, they offer a blueprint for future surveillance capability development in such regions.

Public Health Strategies and Gaps in Madhepura

India's National Action Plan for Rabies Elimination (NAPRE) emphasizes dog vaccination, public education, improved diagnostics, and intersectoral collaboration. In line with this, the following components are crucial for Madhepura and similar areas:

- Vaccination Accessibility: Only a minority of health centers report having regular ARV or RIG stocks. Enhancing cold chain logistics and funding vaccine procurement are urgent needs.
- Community Awareness: Health-seeking behavior is influenced by traditional beliefs and lack of knowledge. Tailored IEC (Information, Education and Communication) campaigns are essential.
- Workforce Training: Primary care providers must be trained in the ID route of ARV administration, wound care, and rabies risk stratification.
- Data Surveillance: Establishing a digital reporting mechanism integrated with IDSP (Integrated Disease Surveillance Programme) will help map incidence trends and plan interventions.

The Promise of "One Health" in Rural Rabies Control

Rabies control demands a unified "One Health" approach that encompasses human health, animal health, and environmental management. In Madhepura, this means:

- Mass dog vaccination campaigns coordinated by local veterinary services.
- Neutering programs to control stray dog populations.
- School and community education targeting children and high-risk groups.
- Partnerships between health departments, NGOs, and local governance.

Such integrative strategies are not only cost-effective but also sustainable in the long term, particularly when bolstered by community engagement.

CONCLUSION

Rabies remains one of the most tragic yet entirely avoidable infectious diseases, disproportionately impacting vulnerable communities across India. Rural areas like Madhepura are particularly affected due to a complex mix of challenges, including inadequate healthcare infrastructure, erratic vaccine availability, poor public knowledge, and a lack of functional diagnostic and surveillance systems. These persistent gaps not only hinder early response but also allow the virus to continue circulating among human and animal populations.

This review underscores the pressing need for well-planned, evidence-driven interventions that move beyond standard protocols. The adoption of innovative molecular diagnostic tools such as RT-LAMP, which have shown success in other low-resource settings, offers significant potential. If deployed effectively, these methods could fill

major voids in rabies detection and monitoring at the local level, enabling quicker diagnosis and more precise control measures.

A critical element of future strategies must be the application of a holistic One Health approach, which recognizes the interconnected nature of human, animal, and environmental health. Coordinated efforts involving large-scale dog immunization, public awareness initiatives, veterinary service enhancement, and stronger healthcare outreach are vital to interrupting the chain of transmission. Additionally, culturally sensitive communication strategies are needed to overcome misconceptions and encourage timely post-exposure prophylaxis (PEP) uptake.

Achieving India's goal of eradicating dog-mediated human rabies by 2030 will demand unwavering political will, consistent investment, integrated planning across sectors, and active participation from local communities. Districts like Madhepura must receive focused attention in terms of funding, resources, and innovative public health solutions to ensure equitable progress.

In summary, the elimination of rabies is within reach, but its success depends on converting global knowledge and national policy into practical, community-specific actions. Strengthening healthcare systems, fostering local awareness, and adopting scalable diagnostic technologies will position India as a global leader in the fight against rabies-related deaths.

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